Photoelectric Effect Exercise

Dr. Carlos Alejandro Trujillo
Modern Physics, Universidad EAFIT

Problem Statement

Suppose that light of total intensity

$$1.0 \ \mu\text{W/cm}^2$$

falls on a clean iron sample of area

$$1.0 \ \text{cm}^2.$$

Assume that the iron sample reflects 96% of the light and that only 3.0% of the absorbed energy lies in the violet region of the spectrum (above the threshold frequency).

1. What intensity is actually available for the photoelectric effect?
2. Assuming that all the photons in the violet region have an effective wavelength of $\lambda = 250$ nm, how many electrons will be emitted per second?
3. Calculate the current in the phototube in amperes.
4. If the cutoff frequency is

   $$f_0 = 1.1 \times 10^{15} \ \text{Hz},$$

   find the work function, ϕ, for iron.
5. Find the stopping voltage for iron if photoelectrons are produced by light with $\lambda = 250$ nm.

Solution

(a) Intensity Available for the Photoelectric Effect

The incident intensity is given by

$$I_{\text{total}} = 1.0\ \mu\text{W/cm}^2 = 1.0 \times 10^{-6}\ \text{W/cm}^2.$$

Since the area of the sample is 1.0 cm^2, the total incident power is

$$P_{\text{incident}} = I_{\text{total}} \times 1.0\ \text{cm}^2 = 1.0 \times 10^{-6}\ \text{W}.$$

Because the sample reflects 96% of the light, only 4% is absorbed:

$$P_{\text{absorbed}} = 0.04 \times P_{\text{incident}} = 0.04 \times 1.0 \times 10^{-6}\ \text{W} = 4.0 \times 10^{-8}\ \text{W}.$$

Finally, since only 3.0% of the absorbed energy is in the violet region:

$$P_{\text{violet}} = 0.03 \times P_{\text{absorbed}} = 0.03 \times 4.0 \times 10^{-8}\ \text{W} = 1.2 \times 10^{-9}\ \text{W}.$$

Thus, the intensity available for the photoelectric effect is:

$$\boxed{1.2 \times 10^{-9}\ \text{W}}.$$

(b) Number of Electrons Emitted per Second

Assume that all photons in the violet region have an effective wavelength of

$$\lambda = 250\ \text{nm} = 250 \times 10^{-9}\ \text{m}.$$

The energy of one photon is given by:

$$E_{\text{photon}} = \frac{hc}{\lambda},$$

where

$$h = 6.626 \times 10^{-34}\,\text{J} \cdot \text{s}, \quad c = 3.0 \times 10^{8}\,\text{m/s}.$$

Plugging in the numbers:

$$E_{\text{photon}} = \frac{6.626 \times 10^{-34}}{250 \times 10^{-9}\ \text{m}} \approx 7.95 \times 10^{-19}\ \text{J}.$$

The number of photons (and hence electrons, assuming one electron per photon) per second is:

$$N = \frac{P_{\text{violet}}}{E_{\text{photon}}} = \frac{1.2 \times 10^{-9}\ \text{W}}{7.95 \times 10^{-19}\ \text{J}} \approx 1.51 \times 10^{9}\ \text{s}^{-1}.$$

Thus, approximately:

$$\boxed{1.5 \times 10^{9}\ \text{electrons/s}}.$$

(c) Photocurrent in the Phototube

Each electron carries an elementary charge of

$$e = 1.602 \times 10^{-19}\ \text{C}.$$

The photocurrent is given by:

$$I = N \times e = 1.5 \times 10^{9}\ \text{s}^{-1} \times 1.602 \times 10^{-19}\ \text{C} \approx 2.403 \times 10^{-10}\ \text{A}.$$

Thus, the photocurrent is:

$$\boxed{2.4 \times 10^{-10}\ \text{A}}.$$

(d) Work Function for Iron

The cutoff frequency is given as:

$$f_0 = 1.1 \times 10^{15}\ \text{Hz}.$$

The work function is:

$$\phi = hf_0 = (6.626 \times 10^{-34}\ \text{J} \cdot \text{s}) \times (1.1 \times 10^{15}\ \text{Hz}) \approx 7.29 \times 10^{-19}\ \text{J}.$$

Converting to electron volts (using 1 eV $= 1.602 \times 10^{-19}$ J):

$$\phi \approx \frac{7.29 \times 10^{-19}\ \text{J}}{1.602 \times 10^{-19}\ \text{J/eV}} \approx 4.55\ \text{eV}.$$

Thus, the work function is:

$$\boxed{7.3 \times 10^{-19}\ \text{J}\ (\approx 4.55\ \text{eV})}.$$

(e) Stopping Voltage

The maximum kinetic energy of the photoelectrons is:

$$K_{\max} = E_{\text{photon}} - \phi \approx 7.95 \times 10^{-19}\ \text{J} - 7.29 \times 10^{-19}\ \text{J} \approx 6.6 \times 10^{-20}\ \text{J}.$$

The stopping voltage V_0 is defined by:

$$eV_0 = K_{\max} \quad \Longrightarrow \quad V_0 = \frac{K_{\max}}{e}.$$

Substituting the values:

$$V_0 = \frac{6.6 \times 10^{-20}\ \text{J}}{1.602 \times 10^{-19}\ \text{C}} \approx 0.41\ \text{V}.$$

Thus, the stopping voltage is:

$$\boxed{0.41\ \text{V}}.$$

Summary of Answers

1. Intensity available for the photoelectric effect: 1.2×10^{-9} W.
2. Number of electrons emitted per second: 1.5×10^{9} electrons/s.
3. Photocurrent: 2.4×10^{-10} A.
4. Work function for iron: 7.3×10^{-19} J (approximately 4.55 eV).
5. Stopping voltage: 0.41 V.